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NTSC SrII optical lattice clock with uncertainty of 2×10^{-18}

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Abstract

We report the systematic evaluation of the SrII optical lattice clock (OLC) at the National Time Service Center, achieving an uncertainty of 1.96×10^{-18}. The newly established ^{87}Sr OLC controls the black-body radiation shift by moving atoms into a temperature-stabilized chamber for clock transition detection. It mitigates the DC Stark shift by using conductive optical films on the inner windows, and reduces both the density shift and AC Stark shift by operating the clock with a relatively low lattice trap depth. Additionally, this clock accommodates changes in the angle between the lattice and gravity directions by rotating the entire setup, allowing for convenient adjustment of the tunneling effect in the shallow lattice.

Keywords: optical lattice clocks, clock uncertainty, spectroscopy, optical lattice

1. Introduction

The optical lattice clock (OLC) has achieved both uncertainty and instability at the 10^{-19} level [1–3] since its first experimental demonstration in 2005 [4]. The exceptional performance of the OLC makes it an excellent candidate for redefining the second in the International System of Units (SI), and the corresponding roadmap is steadily advancing and being completed [5, 6]. One of the key prerequisites before officially adopting the optical second definition is that the agreement of optical frequency ratios between a few (at least 5) other optical frequency standards are found to better than 5×10^{-18}, which necessitates that the systematic uncertainty of each individual clock be below 5×10^{-18}.

For most OLCs, the black-body radiation (BBR) shift and lattice AC Stark shift are the main contributors to systematic uncertainty [1–4, 7–12]. To reduce BBR shift uncertainty, it is crucial to accurately measure the temperature distribution around the atoms using calibrated sensors and create a uniform, well-characterized BBR environment. Common methods include controlling the temperature of the entire science chamber [1, 8, 12] or measuring the clock transition within a BBR shield [7, 13–15]. On the other hand, to minimize the lattice AC Stark shift, it is necessary to measure the shift under various trap depths and lattice light frequencies to determine the shift coefficients relative to the trap depth [16]. Meanwhile, a highly effective approach is to directly reduce the trap depth to the shallow lattice range, can decrease the uncertainty of the lattice AC Stark shift to the low 10^{-19} range [1, 17] and eliminate the density shift at the magic trap depth [18]. However, tunneling between lattice sites in shallow lattices can cause decoherence and potential shifts seen as a results of residual Doppler and recoil effect [19]. Fortunately, these passive effects, reducing the clock stability and accuracy, can be suppressed by aligning the lattice along the gravity direction [17, 19] or using the Floquet-engineering method [20]. However, few OLCs to date have utilized both BBR shield and shallow

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lattice methods, which holds promise for achieving an uncertainty in the low 10^{-19} range.

In this work, we present the new ^{87}Sr OLC at National Time Service Center (NTSC), labeled SrII. Compared with NTSC-SrI [11], SrII integrates several techniques to enhance its performance, including an optical lattice with low trap depth and minimal tunneling effect, a BBR shield with active temperature control at 0 °C, and transparent conductive optical films on the inner surfaces of all windows. These techniques significantly reduce the estimated uncertainties associated with the BBR shift, lattice AC Stark shift, density shift, and DC Stark shift. All known systematic shifts are carefully evaluated, with a thorough discussion of the corresponding uncertainties.

2. Experimental setup

The setup of the SrII is shown in figure 1(a). The entire clock setup is mounted on a rotatable stainless steel board, with the lattice light direction being perpendicular to the rotation axis. The angle between the lattice light and the gravity direction can be adjusted from 0° to 90° using a hydraulic jack. In the science chamber, a BBR shield is positioned 8 mm away from the center of the magneto-optical trap (MOT). The BBR shield is made of copper, with active temperature control provided by a four-stage Peltier cooler and is blackened through high-temperature oxidation. A 1 mm diameter through-hole allows the propagation of the lattice and clock laser beams. A 20 L s^{-1} ion pump and a combination pump (consisting of a 100 L s^{-1} getter pump and a 5 L s^{-1} ion pump) are employed to maintain the vacuum around the atomic dispenser at 4×10^{-9} Pa. Meanwhile, a 100 L s^{-1} getter pump and a combination pump (comprising a 300 L s^{-1} getter pump and a 75 L s^{-1} ion pump) are used to maintain the vacuum in the science chamber at 2×10^{-8} Pa.

When the clock operates, the hot atomic vapor, generated by heating the atomic dispenser, passes through stainless steel capillary tubes and undergoes two-dimensional optical alignment to reduce its divergence angle. After being cooled by a Zeeman slower, the atoms reach the science chamber, where they are trapped and cooled by the blue MOT using the $|5s^2\ ^1S_0(F=9/2)\rangle \rightarrow |5s5p\ ^1P_1(F=11/2)\rangle$ transition at 461 nm. To increase cooling efficiency, repumping lasers are used: the $|5s5p\ ^3P_0\rangle \rightarrow |5s6s\ ^3S_1\rangle$ transition at 679 nm and $|5s5p\ ^3P_2\rangle \rightarrow |5s6s\ ^3S_1\rangle$ transition at 707 nm repump atoms that decay to the triplet levels ($|5s5p\ ^3P_0\rangle$ and $|5s5p\ ^3P_2\rangle$) back to the ground state ($|5s^2\ ^1S_0\rangle$) [21]. Before the blue MOT concludes, the trap laser power is linearly reduced to zero over a 30 ms period to minimize atomic temperature. Once the blue MOT phase ends, the magnetic field strength is reduced to 3 G cm^{-1}, and the 679 nm, 707 nm, and 461 nm lasers are turned off. The red MOT then operates to further cool the atoms to approximately 3 μK, utilizing the transitions $|5s^2\ ^1S_0(F=9/2)\rangle \rightarrow |5s5p\ ^3P_1(F=11/2)\rangle$ and $|5s^2\ ^1S_0(F=9/2)\rangle \rightarrow |5s5p\ ^3P_1(F=9/2)\rangle$, both at 689 nm [22].

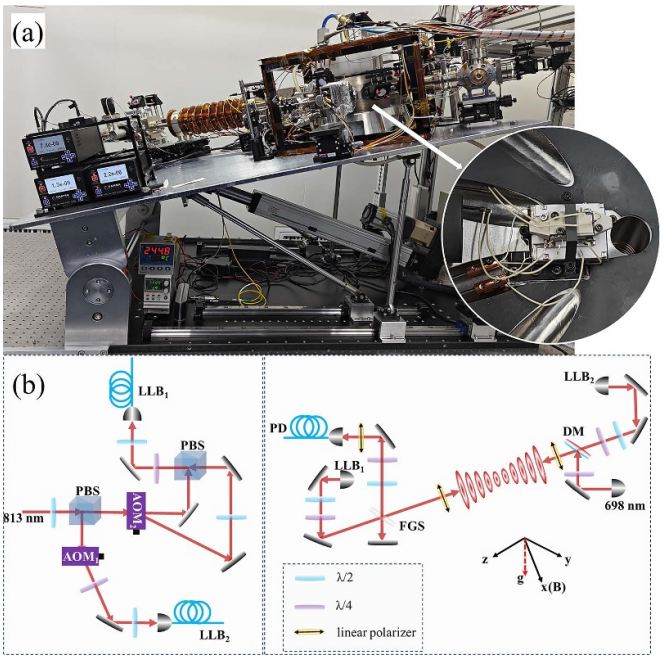


Figure 1. The setup of the SrII in NTSC. (a) Overview of the setup. The inset highlights the science chamber and its internal structure. The inner surfaces of the science chamber are covered with anodized-black stainless steel sheets to reduce diffuse reflection. A copper BBR shield is positioned on the cooling surface of a four-stage Peltier cooler to regulate its temperature. A PT1000 and two PT100 platinum resistors are used to sample the temperature of the BBR shield. Polyether-ether-ketone (PEEK) plates are placed on the shield's surfaces to prevent heat exchange with the surrounding environment. (b) Setup of the lattice and clock transition detection. The optical lattice is created by overlapping two 813 nm lattice light beams (LLB_1 and LLB_2), which are frequency-shifted by the acousto-optic modulator (AOM). The minus first-order and zero-order diffracted light from LLB_1 are combined using a polarized beam splitter (PBS), then coupled into a single fiber. About 4% of the power from both LLBs is reflected by a flat glass sheet (FGS), and the minus first-order light from LLB_1 is suppressed before reaching the photodiode (PD) by adjusting the linear polarizer and $\lambda/4$ and $\lambda/2$ phase retarders. The relative phase between the two LLBs is locked based on the PD-detected phase variation that is used to feedback the frequency of the minus first-order light of LLB_2. Since the polarization of the minus first-order and zero-order diffracted light from LLB_1 are orthogonal, the zero-order light is blocked from entering the science chamber when the minus first-order light passes maximally through the linear polarizer. There is a 75° angle between the gravitational direction and the lattice's longitudinal direction when the clock is operating regularly. The 698 nm clock laser is overlapped with the lattice light beams using a dichroic mirror (DM), where fiber phase noise of the clock laser is suppressed by reflecting 20% of the light power from the fiber's output face.

The lattice lasers are continuously illuminated, naturally loading atoms into the one-dimensional optical lattice formed by two opposite-propagating lattice laser beams. The lattice laser is generated by a fiber laser with a maximum output power of 2.4 W. It is spectrally filtered using an interference filter with a bandwidth of 0.2 nm, and its frequency

is stabilized to the optical frequency comb (OFC). The carrier-envelope offset frequency ($f_{\rm ceo} \sim 50$ MHz) and repetition frequency ($f_{\rm r} \sim 200$ MHz) of the OFC are referenced to the clock laser, whose phase is stabilized using a 20 cm-long ultralow-expansion cavity with a finesse of 200 000. The atoms are then spin-polarized into the Zeeman sublevels $|m_{\rm F} = \pm 9/2\rangle$ using $\sigma^{\pm}$-polarized light, which corresponds to $|5s^2\ {}^1S_0(F = 9/2)\rangle \rightarrow |5s5p\ {}^3P_1(F = 9/2)\rangle$ transition [23].

Atoms are initially loaded into the lattice with a trap depth of $U = 124\ E_{\rm R}$ (where $E_{\rm r}$ is the photon recoil energy of the lattice light). The trap depth is then linearly reduced to $40\ E_{\rm R}$ over 5 ms to remove hotter atoms [23]. After holding this depth for 10 ms, the trap depth is linearly increased to the required depth for clock operation. This process reduces the atomic temperature to below 2 μK, as indicated by the sideband spectra shown in figure 2(a) [24]. Before the clock transition interrogation, atoms are moved into the BBR shield by linearly increasing the frequency of one of the lattice lights by 1 MHz over 5 ms [25]. The frequency deviation is held for 40 ms and then decreased to zero within 5 ms. This procedure displaces the atoms by 17.29 mm, with the atoms being 10.5(2) mm away from the hole output (with a diameter of 1.00(3) mm), as inferred from the lattice image and hole positions. After the clock transition interrogation, atoms are moved out of the BBR shield and detected the excitation fraction. The trap depth decreases from 124 $E_{\rm R}$ to 71.6 $E_{\rm R}$ as the atoms move from their initial position to the BBR shield, indicating that the effective lattice light intensity at the BBR shield is reduced by a factor of 0.58 compared to the initial position.

Figure 2(b) shows the single clock instability using the interleaved self-comparison method [12, 26, 27]. The two interleaved clocks operate with identical systematic parameters, and the frequency difference between them is divided by a factor of 2 to estimate the single clock instability [12, 26]. The extracted clock stability is $3.6 \times 10^{-16}\ (\tau/{\rm s})^{-0.5}$, which is slightly worse than the Dick limit of $2.8 \times 10^{-16}\ (\tau/{\rm s})^{-0.5}$ inferred from the flicker noise floor of clock laser (4×10^{-16}) [28].

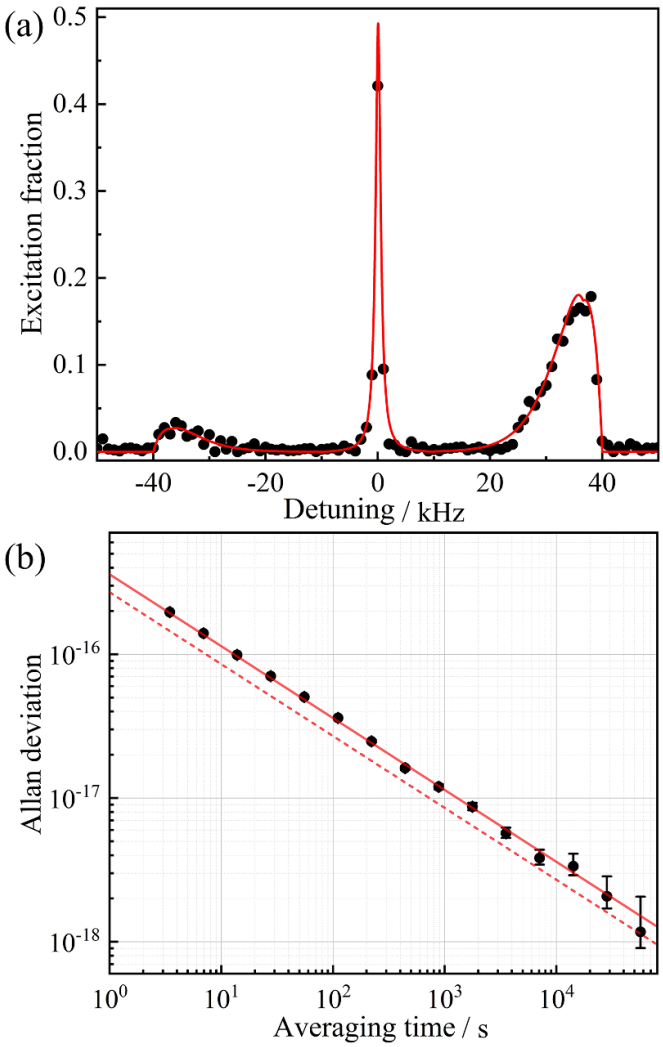


Figure 2. (a) Sideband spectra. The curve represents a fit of the sideband line shape. Based on the fitting results, the longitudinal and radial atomic temperatures are $T_{\rm z} = 1.2$ μK and $T_{\rm r} = 1.9$ μK, respectively. (b) The single clock instability determined by the interleaved self-comparison method. The solid line represents a linear fit of the data with a constant slope of −0.5, while the dotted line indicates the corresponding Dick-limited instability, assuming a 0.4 s Rabi pulse and a clock cycle of 0.865 s.

3. Systematic shifts of the ^{87}Sr clock transition

To ensure consistency among identical optical clocks, all known systematic shifts must be characterized and corrected. A summary of the systematic shifts and their corresponding uncertainty contributions is provided in table 1.

3.1. BBR shift

The BBR shift originates from the differential Stark shift between the clock states, which is caused by the interaction of strontium atoms with the BBR field emitted by the surrounding environment. It is typically the dominant frequency correction and source of systematic uncertainty in state-of-the-art OLCs. The Stark shift of the BBR electric field characterized by the temperature T can be written as [29, 30]

$$\begin{aligned}\Delta\nu_{\rm BBR} &= -\frac{1}{2h}\int_0^{\infty} \Delta\alpha_{eg}(\omega)\, E^2(T,\omega)\, {\rm d}\omega \\ &= \Delta\nu^{({\rm stat})}(T) + \Delta\nu^{({\rm dyn})}(T). \end{aligned} \quad (1)$$

The static BBR shift $\Delta\nu^{({\rm stat})}(T)$, is given by $\nu_{\rm stat}(T/T_0)^4$ with $\nu_{\rm stat} = -2.130\,23(6)$ Hz at $T_0 = 300$ K [30]. And the remaining dynamic correction term of the BBR shift $\Delta\nu^{({\rm dyn})}(T)$, can be represented by a higher-order polynomial

Table 1. SrII systematic uncertainty. All values are in units of Hz/Hz.

Systematic effect	Correction (10^{-18})	Uncertainty (10^{-18})
BBR	−3608.59	0.7
Lattice AC Stark	−0.41	1.56
Density	−5.43	0.48
Background gas	−6.38	0.64
1st order Zeeman	0	<0.1
2nd order Zeeman	−86.68	0.11
First order Doppler	0	0.5
DC Stark	0	<0.1
Tunneling	0	<0.1
Line pulling	0	<0.1
Servo error	0	0.12
Probe AC Stark	0	<0.1
AOM phase chirp	0	<0.1
Total	−3707.49	1.96

expansion of

$$\Delta\nu^{(\mathrm{dyn})}(T) \approx \eta_6\left(\frac{T}{T_0}\right)^6 + \eta_8\left(\frac{T}{T_0}\right)^8 + \eta_{10}\left(\frac{T}{T_0}\right)^{10} + \eta_{12}\left(\frac{T}{T_0}\right)^{12} + \eta_{14}\left(\frac{T}{T_0}\right)^{14} + \eta_{16}\left(\frac{T}{T_0}\right)^{16}. \quad (2)$$

Using the latest data for the lifetime of the $|5s4d\,^3D_1\rangle$ state [1], as well as experimental data such as the 813 nm magic wavelength, 390 nm magic wavelength, turnout frequency and static polarizability, we re-evaluated the BBR shift based on the self-consistent method described in [1, 29]. A set of experimental measurements was selected as the fitting target dataset. In the theoretical calculations of these observables, the Einstein A coefficients of several transitions were treated as adjustable parameters. These coefficients were varied within their uncertainty bounds to optimize the fit to the experimental data using the least-squares method [31]. Finally, using the optimized A coefficients from the fitting procedure, we calculated the dynamic frequency shift $\Delta\nu^{(\mathrm{dyn})}(T)$ as a function of temperature T, spanning from 250 K to 320 K (see figure 3(a)).

To comprehensively assess the uncertainty of $\Delta\nu^{(\mathrm{dyn})}(T)$, we employed Monte Carlo sampling. A set of experimental observables x_i were randomly drawn from their distributions (e.g. $x_i \sim \mathcal{N}(\mu_\mathrm{m}, \sigma_\mathrm{m})$, where $\mathcal{N}(\mu_\mathrm{m}, \sigma_\mathrm{m})$ represents a Gaussian distribution with mean μ_m and standard deviation σ_m). The sampled experimental data were then applied in a least-squares fitting procedure to calculate $\Delta\nu^{(\mathrm{dyn})}(T)$. This process was repeated 10 000 times, and the resulting distribution enabled the estimation of the uncertainty in $\Delta\nu^{(\mathrm{dyn})}(T)$. After incorporating the inflation factor from the leave-one-out method described in [1], the dynamic BBR shift was determined to be $\Delta\nu^{(\mathrm{dyn})} = -153.13(27)$ mHz at 300 K [31], which is in good agreement with the value of $-153.06(33)$ mHz reported in [1].

By fitting the numerical results in the temperature range from 50 K to 350 K using equation (2), the coefficients η_6, η_8, η_{10}, η_{12}, η_{14} and η_{16} are determined to be $-0.131\,22(26)$ Hz, $-0.019\,2904(16)$ Hz, $0.005\,0429(43)$ Hz, $-0.009\,7581(53)$ Hz, $0.001\,3046(31)$ Hz, and $0.000\,700\,06(71)$ Hz, respectively. The maximum deviation between the fitting and numerical results is less than 0.03 mHz.

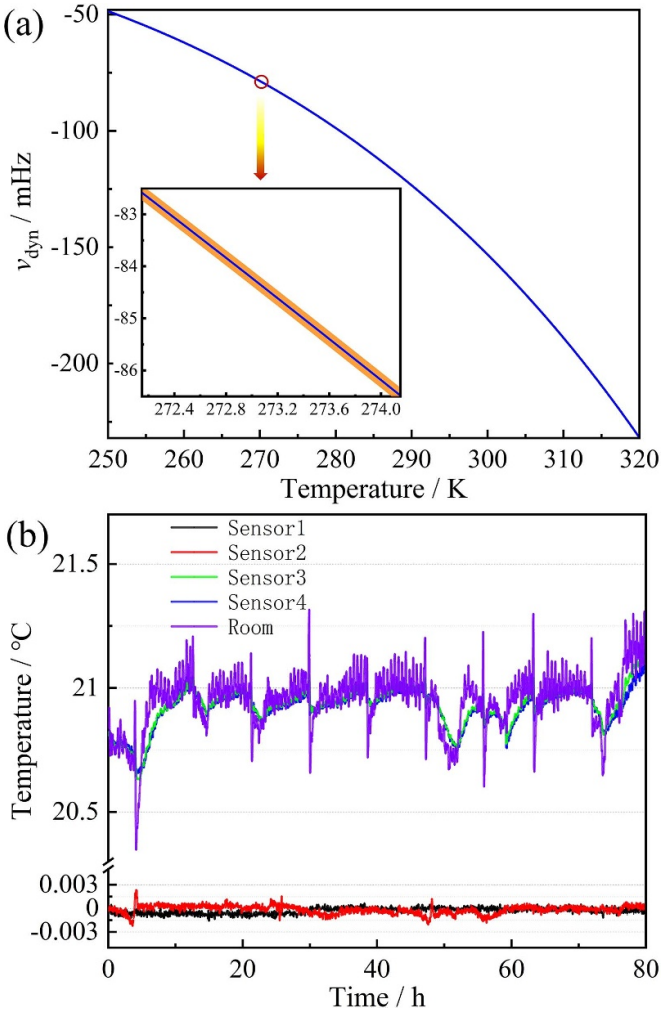


Figure 3. (a) The numerical results of the dynamic BBR shift as a function of temperature. The inset shows the details around the operational temperature of 273.15 K. The orange band indicates the 1σ confidence interval. (b) The temperature data of all sensors. Sensor1 and Sensor2 correspond to the two PT100 platinum resistors located in the BBR shield, with Sensor1 positioned near the MOT side and Sensor2 near the window side. Sensor3 and Sensor4 are placed on the two windows through which the lattice light beams propagate.

In our system, the clock transition interrogation occurs within the BBR shield, located 10 mm from the surface closest to the MOT and 20 mm from the opposite surface. The average temperature of the BBR shield is stabilized at $-4.4(1) \times 10^{-4}$ °C (273.150 44(1) K). The inner surface of the BBR shield is blacked using a high-temperature oxidation method, yielding an emissivity of 0.85(5). Considering an effective solid angle of $3.997\,52(16)\pi$ sr, with uncertainty contributions

Table 2. The BBR shift. All values are in units of Hz/Hz.

Items	Shift (10^{-18})	Uncertainty (10^{-18})
BBR shield	−3605.60	0.37
Room temperature	−2.99	0.11
Temperature uncertainty	0	0.56
multiple reflections	0	0.18
Total	−3608.59	0.7

from the emissivity (78.6%), atom position (20.9%) and hole diameter (0.5%), the BBR shield results in a relative BBR shift of $-3605.60(37) \times 10^{-18}$. Radiation from the room temperature ($T_{\text{room}} = 20.9(5)$ °C) can leak into the BBR shield through the two apertures, corresponding to a solid angle of $2.92(9)\pi$ msr. This results in a BBR shift of -2.99×10^{-18} with an uncertainty of 1.1×10^{-19}. The thermometers are calibrated within 10 mK, the reproducibility of the controllers is evaluated to 2 mK, and the potential temperature gradient within the BBR shield is estimated to be below 1 mK. These uncertainties contribute to a BBR shift uncertainty of 5.6×10^{-19}. Using Monte Carlo ray-tracing simulations, we calculate the BBR shift due to multiple reflections from the inner walls of the BBR shield to be 1.8×10^{-19}, based on an emissivity of 0.8(1) for the anodized-black stainless steel sheets, and we consider this as the systematic uncertainty. As shown in table 2, the total BBR shift is determined to be -3608.59×10^{-18} with an uncertainty of 0.7×10^{-18}.

3.2. Density shift

The density shift in OLCs has been extensively studied since its first experimental observation in the ^{87}Sr OLC [32]. In the context of a spin-polarized cold fermion ensemble, the density shift is typically dominated by p-wave interactions and is linearly dependent on the atomic density [33, 34]. A common method to quantify the density shift is the self-comparison technique, where the atom number is modulated interleaved without altering the distribution of motional states. The shift between two interleaved-operating clocks is directly proportional to the difference in atom number. Recently, *iu-situ* measurements between different regions of a lattice have been shown to provide high accuracy for density shift measurements [18, 25, 35], free from clock laser noise. However, this method requires careful calibration to account for potential errors arising from gradients in ambient temperature, magnetic fields, and the gains of the electron-multiplying charge-coupled device. Therefore, we use the interleaved method to measure the density shift by modulating the operational time of the blue MOT.

To minimize potential errors due to variations in excitation fraction, atomic temperature, and trap depth, we perform nine measurements over two weeks [8], as shown in figure 4. With an operational trap depth of 39.8 E_{R} and an average atom number of 1000, the density shift is measured as -5.43×10^{-18} with an uncertainty of 0.48×10^{-18}. While the absolute atom number has a larger uncertainty of about 20% limited by the knowledge of effective solid angle and detection laser intensity, this does not contribute to the uncertainty in the density shift measurement since the measurements and corrections are based on the voltage signal of the photomultiplier tube.

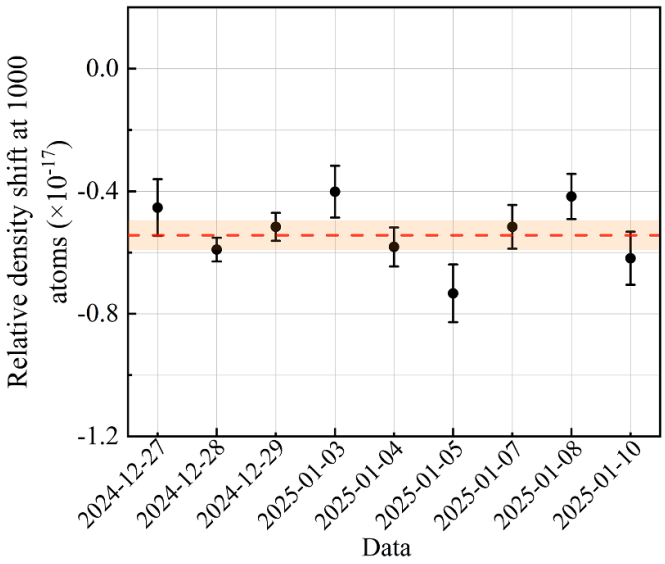


Figure 4. Density shift evaluations relative to the clock transition frequency ν_{sr}. Measurements were performed over two weeks at a trap depth of 39.8 E_{R}. Dashed line is the weighted mean of all measurements. The orange band is the 1σ confidence interval in the weighted mean of these data. The confidence interval has been inflated by the square root of $\chi^2 = 1.73$.

3.3. Lattice AC Stark shift

The lattice laser can induce a differential AC Stark shift for the clock transition, which can be suppressed by operating the lattice light at a wavelength near the magic wavelength. Our lattice laser is based on sum-frequency generation and fiber amplification (Shanghai Precilaser Ltd). The potential shift caused by background spontaneous radiation is minimized to below the 2×10^{-19} level, thanks to the use of a 0.2 nm bandpass filter in our system [36]. Using the thermal model method [8, 37], the lattice AC Stark shift can be expressed as

$$\frac{\Delta\nu_{\text{L}}}{\nu_{\text{sr}}} = -\frac{\partial\alpha^*}{\partial\nu_{\text{L}}}(\nu_{\text{L}} - \nu_{\text{zero}})\,U - \beta^* U^2, \tag{3}$$

where ν_{L} is the lattice frequency, ν_{zero} represents the magic frequency, α^* and β^* are experiment-specific coefficients.

We measured the differential AC Stark shift between U and reference trap depth $U_{\text{ref}} = 39.8$ E_{R} at different lattice frequencies ν_{L}: 368 554 384.1 MHz, 368 554 424.1 MHz, 368 554 472.3 MHz, 368 554 512.3 MHz and 368 554 552.3 MHz, using the interleaved self-comparison method, as shown in figure 5(b). Trap depth-dependent density shifts are systematically removed from the lattice AC Stark

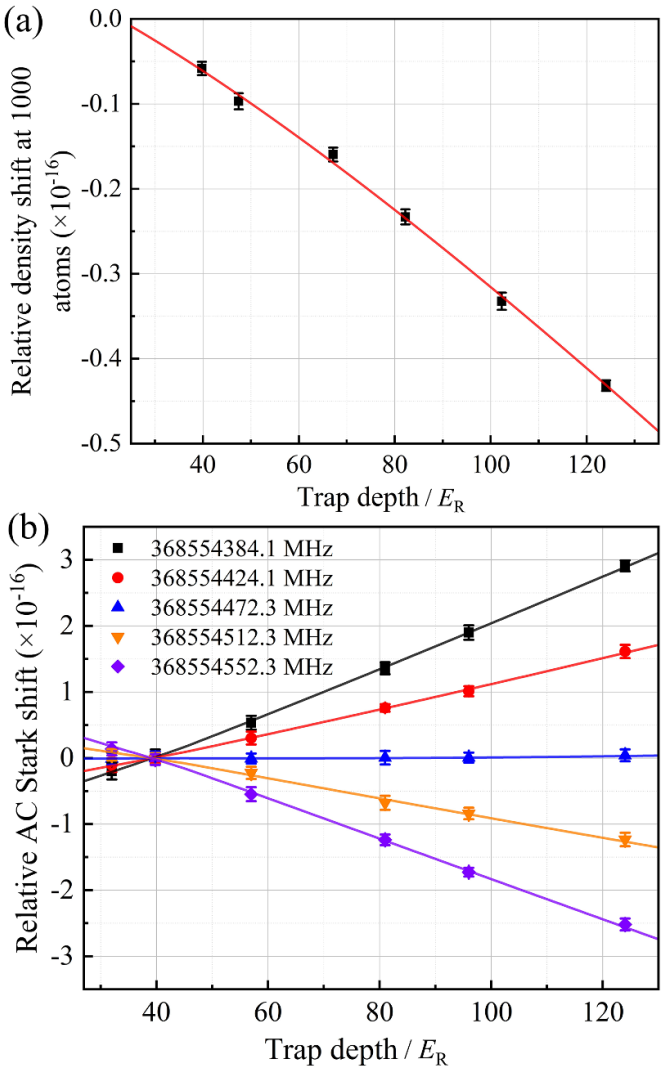


Figure 5. (a) Relative density shifts as a function of trap depth U. The solid line indicates a fit to the data with the expected scaling of the shift as $U^{5/4}$ (b) differential lattice AC Stark shifts measured relative to a trap depth of 39.8 E_R. Curves indicate independently fit using based on equation (3).

shift evaluations point-by-point using density shift measurements across a range of lattice depths, as shown in figure 5(a). The density shift measurements are fitted with a $U^{5/4}$ potential model, and the resulting fitting uncertainty is combined in quadrature with the statistical uncertainty of each lattice light shift measurement. The experimental data of figure 5(b) were fitted using equation (3), where an offset was added since the data represent the differential shift between two trap depths. From the fitting, we obtained $\partial\alpha^*/\partial\nu_L = 33.45(15) \times 10^{-21}\,\mathrm{MHz}^{-1}\,E_R^{-1}$, $\beta^* = -2.6(34) \times 10^{-22}\,E_R^{-2}$, and $\nu_{zero} = 368\,554\,468.2(11)$ MHz. By adjusting the operating lattice frequency to 368 554 468.2 MHz, the lattice AC Stark shift was determined to be -0.41×10^{-18} with an uncertainty of 1.56×10^{-18} at the operational trap depth of 39.8(5) E_R.

Table 3. The uncertainty of the lattice AC Stark shift. All values are in units of Hz/Hz.

Parameter	Uncertainty (10^{-18})
$\frac{\partial\alpha^*_{\nu_L}}{\partial\nu_L}$	<0.01
β^*	0.54
ν_{zero}	1.46
U	0.021
Total	1.56

Table 3 summarizes the uncertainty budget for the lattice AC Stark shift evaluation.

3.4. Background gas collision shift

The residual gas in the science chamber can lead to a frequency shift similar to the density shift. Assuming that the background gas is primarily hydrogen molecules, the background gas collision shift is given by $-3.0(3) \times 10^{-17}/\tau_b$ [38], where τ_b is the lattice trap lifetime in a unit of second. For our system, the lattice trap lifetime is measured to be 4.7(1) s, which corresponds to a background gas collision shift of -6.38×10^{-18} with an uncertainty of 0.64×10^{-18}.

3.5. Zeeman shift

The clock laser is locked to the average frequency of $|m_F = \pm 9/2\rangle \rightarrow |m_F = \pm 9/2\rangle$ transitions, ensuring that the 1st-order Zeeman shift is canceled. However, any drift in the magnetic field can induce a residual 1st-order Zeeman shift, which can be expressed as [12]

$$\Delta\nu_{Z1} \leqslant \langle \Delta S_{2n} - \Delta S_{2n-1} \rangle / 8, \tag{4}$$

where $\langle \Delta S_{2n} - \Delta S_{2n-1} \rangle$ represents the average drift rate of the Zeeman splitting (labeled by ΔS) between $|m_F = \pm 9/2\rangle$ across the $2n$ and $2n-1$ clock cycles. Based on the in-loop operation data, the residual first-order Zeeman shift is estimated to be smaller than 5×10^{-20}.

The 2nd-order Zeeman shift can be expressed as [8]

$$\Delta\nu_{Z2} = \beta_{z2} \Delta S^2, \tag{5}$$

where the coefficient β_{z2} has been accurately determined to be $-2.456(3) \times 10^{-7}\,\mathrm{Hz}^{-1}$ [8]. Although the 2nd-order Zeeman shift is dynamically corrected based on the ΔS measured in real time, the typical value of $\Delta S = 389.21$ Hz corresponds to a 2nd-order Zeeman shift of -86.68×10^{-18}, with an uncertainty of 0.11×10^{-18}.

3.6. First order Doppler shift

The relative phase between the two lattice lasers is actively stabilized, as shown in figure 1(b). The residual phase variation between the two beams arises from the uncompensated path

of $L_1 \approx 12$ cm between the -1st-order and zero-order lights of LLB_1. There are also two uncompensated paths: $L_2 \approx 10$ cm between lattice and the clock lasers, and $L_3 \approx 35$ cm between the clock laser output and the reference face of the fiber phase noise compensation. These path lengths introduce an uncertainty in the first-order Doppler shift of 0.5×10^{-18}, conservatively assuming a temperature variation rate of 1 K h^{-1} [39].

3.7. DC Stark shift

All surfaces enclosed in the vacuum of the viewports are coated with indium tin oxide film. The transmissivity of the windows is 90%, 93% and 95% for the 461 nm, 689 nm and 813 nm wavelengths, respectively. The electrical resistance measurements show that both the BBR shield's inner-surface oxidation layer (<5 kΩ) and the anodized-black stainless-steel sheets (<0.2 kΩ for the anodized layer) provide adequate conductivity to avoid stray charge accumulation on their surfaces. By grounding the chamber, which simultaneously grounds the BBR shield and the anodized-black stainless-steel sheets, the DC Stark shift is suppressed below the 0.1×10^{-18} level through the implementation of an in-vacuum Faraday shield [15].

3.8. Tunneling

The 75° angle between the lattice's longitudinal direction and gravity breaks the translational symmetry and engineers the Wannier–Stark (WS) ladder potential [1, 17–19]. In this scenario, if the atomic ensemble, after cooling, resides in a thermal state (an incoherent superposition of WS states across different lattice sites), the carrier transition will not experience a frequency shift due to off-resonant off-site excitations. However, if the initial state of the system includes coherent superpositions of WS states from neighboring wells, these components will interfere during clock laser interrogation, leading to a frequency shift proportional to $\Omega_0\Omega_1/\Delta_g$ [19], where Ω_0 and Ω_1 represent the Rabi frequencies of the carrier and the 1st-order sidebands, and Δ_g is the frequency difference between adjacent sidebands. In our system, $\Omega_0/2\pi = 1.46(6)$ Hz, $\Omega_1/2\pi = 1.15(5) \times 10^{-4}$ Hz [18] and $\Delta_g/2\pi = 226$ Hz, which corresponds to a tunneling-induced shift of 0.011×10^{-18}.

3.9. Other shifts

Off-resonant excitation due to unpurified state preparation will induce line pulling to the $|m_F = \pm 9/2\rangle \rightarrow |m_F = \pm 9/2\rangle$ transitions. We have quantified that the population of the $|m_F| = 7/2$ state is less than 4%. For a 0.4 s Rabi pulse and a 39.8 Hz Zeeman splitting, the maximum off-resonant excitation fraction is determined to be 6.6×10^{-5}, derived from the thermally averaged Rabi spectrum [24] and the squared Clebsch–Gordan coefficient ratio of 0.6 [8]. This excitation variation induces a frequency shift of 4.6×10^{-21}. However, averaging over the stretched states $|m_F = \pm 9/2\rangle$ suppresses this shift. Therefore, we consider the above value as the uncertainty arising from the line-pulling effect.

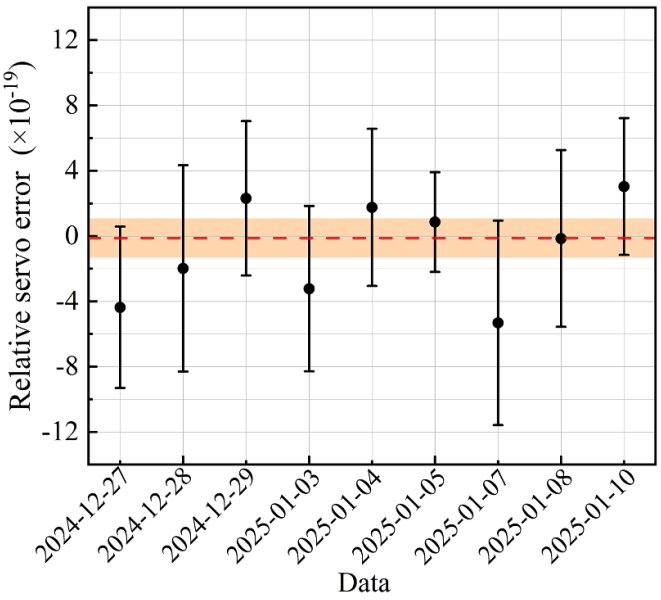


Figure 6. Measurements of the servo error. The servo errors were determined by averaging the lock error signal recorded during density shift measurements. The error bars represent the penultimate point of the Allan deviation from the corresponding dataset. The red dashed line indicates the weighted mean of all measurements, yielding a reduced chi-squared value of 0.83. The orange band represents the 1σ confidence interval for the weighted mean of these data.

The clock laser frequency drift is canceled by a second integration stage with a servo time constant of 50 s using the lock error signal acquired during clock operation. However, the residual frequency drift due to the long integration time potentially leads to a servo error, as the frequency compensation always lags behind the clock laser frequency deviation. The servo error can be determined by averaging the lock error signal displayed in figure 6. Based on nine measurements conducted over a two-week period, the servo error is estimated at 1.6×10^{-20} with an uncertainty of 0.12×10^{-18}.

The probe light itself induces an AC Stark shift, which is directly proportional to the square of Ω_0. Based on the accurately measured coefficient of $-6.0(5) \times 10^{-7}\,\mathrm{Hz}^{-1}$ for the ^{87}Sr OLC [40], the probe Stark shift is determined to be $-3.0(3) \times 10^{-21}$.

At the end of the clock transition interrogation, the clock laser frequency is detuned by 1 MHz from the resonance frequency. This method not only maintains the lock states of the fiber phase noise cancellation and the clock laser intensity stabilization but also avoids thermally induced differential path length transients in the acousto-optic modulator (AOM) crystal. Thus, the effect caused by the AOM phase chirp can be neglected.

Based on the above evaluation results, the uncertainty of NTSC-SrII OLC is 1.96×10^{-18}.

4. Conclusion

In summary, we have developed a ^{87}Sr OLC with a clock instability of $3.6 \times 10^{-16}\ (\tau/\mathrm{s})^{-0.5}$ and a systematic uncertainty of 1.96×10^{-18}. The current dominant sources of systematic uncertainty are the lattice AC Stark shift, background collisions, and the density shift. In the future, we plan to improve the vacuum conditions in the science chamber and further reduce the trap depth to mitigate the lattice AC Stark shift and density effects. Additionally, we aim to measure the absolute frequency of our ^{87}Sr OLC and contribute to the generation of international atomic time.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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